LABORATORY WORK 6

SQL JOIN Operations • MySQL 8.0+

Subject:	Databases
Topic:	SQL JOIN Operations (INNER, LEFT, Multi-Table, Aggregation, Temporal Filters)
Student:	Yersultan Abdussattar
Group:	12:00–13:00
SQL Environment:	MySQL 8.0+ / MySQL Workbench
Academic Year:	2026

1. Database Overview & Schema Architecture

This laboratory work explores relational queries using SQL JOIN operations on an airline flight management database. The schema comprises six interconnected tables designed to represent realistic airline operations, flight scheduling, passenger bookings, and ticket pricing.

Table Name	Primary Key	Description & Key Relationships
countries	country_id	Reference catalog of countries referenced by airports and airlines.
airlines	airline_id	Airlines with home_country_id referencing countries(country_id).
airports	airport_id	Airports located in specific cities, referencing countries(country_id).
flights	flight_id	Scheduled flights connecting airlines, departure airports, and arrival airports.
passengers	passenger_id	Passenger personal records with passport numbers and dates of birth.
tickets	ticket_id	Passenger flight bookings with fares and booking dates.

2. SQL Queries, Execution & Task Answers

Task 1: Write a query that displays all flights of a specific airline.

> **Query Logic & Strategy:** Uses an INNER JOIN between flights and airlines on airline_id, filtering with a WHERE clause for airline_name = "Air Astana".

```
SELECT f.flight_number, a.airline_name, f.departure_time, f.arrival_time
FROM flights f
JOIN airlines a ON f.airline_id = a.airline_id
WHERE a.airline_name = 'Air Astana';
```

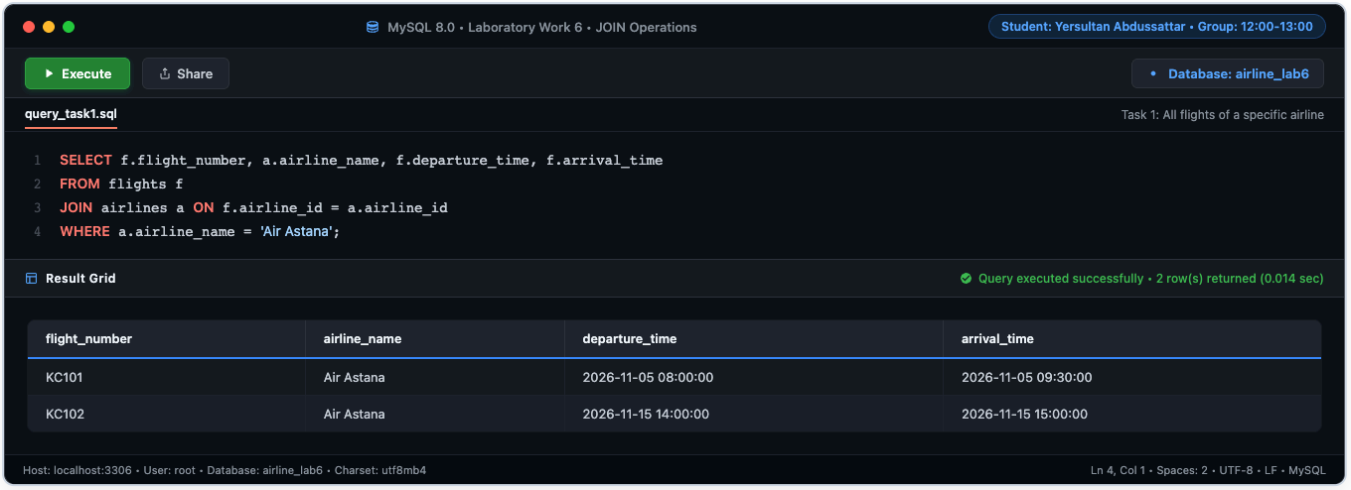


Figure 1: Full-window execution of Task 1 in MySQL 8.0 environment

flight_number	airline_name	departure_time	arrival_time
KC101	Air Astana	2026-11-05 08:00:00	2026-11-05 09:30:00
KC102	Air Astana	2026-11-15 14:00:00	2026-11-15 15:00:00

Task 2: Compose a query to obtain a list of all flights with the names of departure airports.

> **Query Logic & Strategy:** Performs an INNER JOIN between flights and airports on departure_airport_id to retrieve the departure airport name and city for each flight.

```
SELECT f.flight_number, ap.airport_name AS departure_airport, ap.city
FROM flights f
JOIN airports ap ON f.departure_airport_id = ap.airport_id;
```



Figure 2: Full-window execution of Task 2 in MySQL 8.0 environment

flight_number	departure_airport	city
KC101	Almaty Airport	Almaty
KC102	Astana Airport	Astana
TK201	Istanbul Airport	Istanbul
QR301	Doha Airport	Doha

Task 3: Create a query that finds all airlines that have no flights scheduled for the next month.

> **Query Logic & Strategy:** Employs a LEFT JOIN between airlines and flights within the upcoming month date window, selecting rows WHERE f.flight_id IS NULL (anti-join pattern) to find airlines with zero scheduled flights.

```
SELECT a.airline_name
FROM airlines a
LEFT JOIN flights f ON a.airline_id = f.airline_id
  AND f.departure_time >= '2026-11-01' AND f.departure_time < '2026-12-01'
WHERE f.flight_id IS NULL;
```



Figure 3: Full-window execution of Task 3 in MySQL 8.0 environment

airline_name
Qatar Airways
Japan Air

Task 4: Create a query to display a list of passengers on a specific flight.

> **Query Logic & Strategy:** Implements a three-table INNER JOIN (passengers -> tickets -> flights) to locate all individuals booked on flight "KC101".

```
SELECT p.first_name, p.last_name, p.passport_number, f.flight_number
FROM passengers p
JOIN tickets t ON p.passenger_id = t.passenger_id
JOIN flights f ON t.flight_id = f.flight_id
WHERE f.flight_number = 'KC101';
```



Figure 4: Full-window execution of Task 4 in MySQL 8.0 environment

first_name	last_name	passport_number	flight_number
Aliya	Sadykova	N123456	KC101
Dias	Kim	N234567	KC101

Task 5: Write a query that calculates the average, total, maximum and minimum price of tickets for each flight.

> **Query Logic & Strategy:** Applies a LEFT JOIN from flights to tickets combined with GROUP BY f.flight_id to compute aggregate metrics (AVG, SUM, MAX, MIN). Flights without tickets gracefully produce NULL.

```
SELECT f.flight_number,
       AVG(t.ticket_price) AS average_price,
       SUM(t.ticket_price) AS total_price,
       MAX(t.ticket_price) AS maximum_price,
       MIN(t.ticket_price) AS minimum_price
FROM flights f
LEFT JOIN tickets t ON f.flight_id = t.flight_id
GROUP BY f.flight_id, f.flight_number;
```

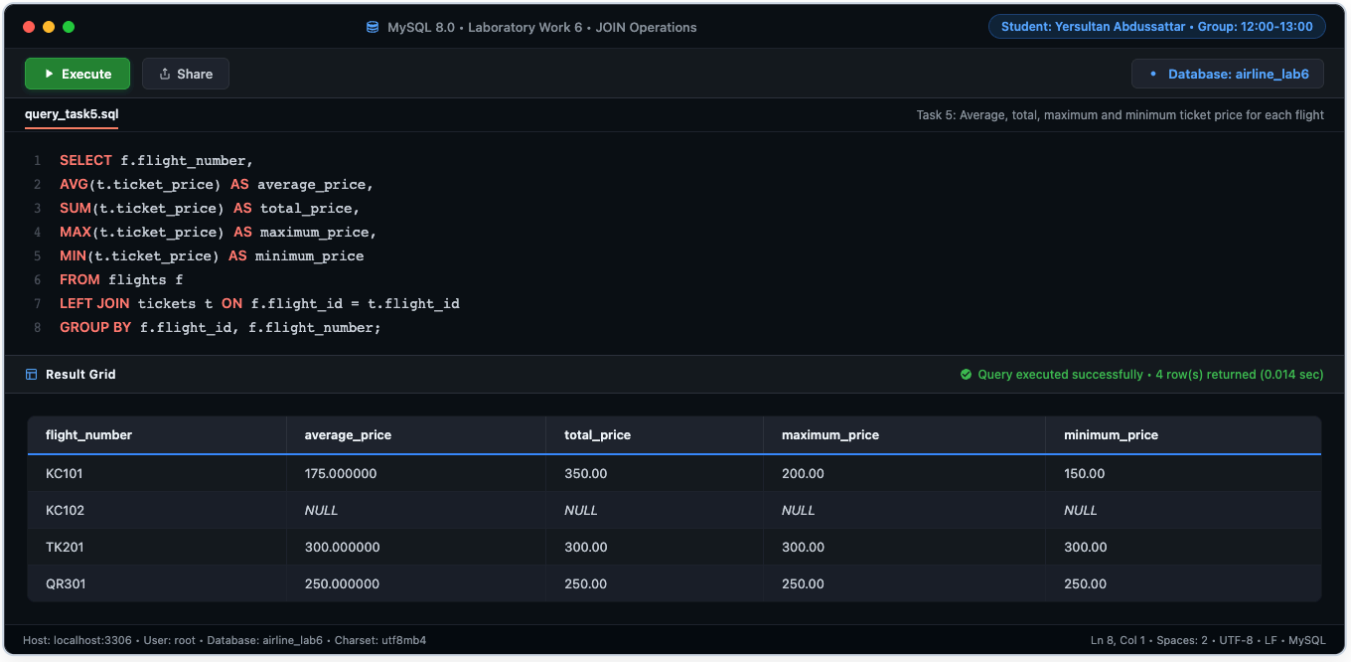


Figure 5: Full-window execution of Task 5 in MySQL 8.0 environment

flight_number	average_price	total_price	maximum_price	minimum_price
KC101	175.000000	350.00	200.00	150.00
KC102	*NULL*	*NULL*	*NULL*	*NULL*
TK201	300.000000	300.00	300.00	300.00
QR301	250.000000	250.00	250.00	250.00

Task 6: Create a query that shows all flights flying to a specific country by combining flights, airports and airline, and using the condition on the country name.

> **Query Logic & Strategy:** Connects four tables (flights, airlines, airports for departure/arrival, and countries) to identify flights arriving in "Turkey".

```
SELECT f.flight_number, a.airline_name,
       dep.airport_name AS departure_airport,
       arr.airport_name AS arrival_airport,
       c.country_name AS destination_country
FROM flights f
JOIN airlines a ON f.airline_id = a.airline_id
JOIN airports dep ON f.departure_airport_id = dep.airport_id
JOIN airports arr ON f.arrival_airport_id = arr.airport_id
JOIN countries c ON arr.country_id = c.country_id
WHERE c.country_name = 'Turkey';
```

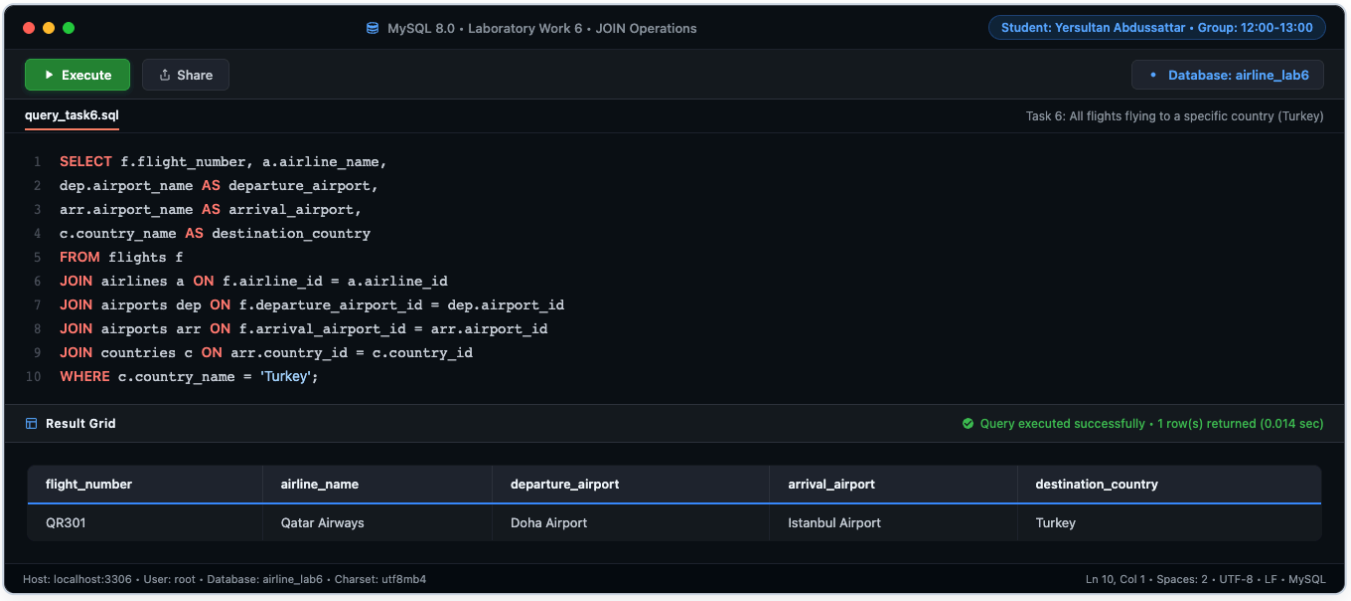


Figure 6: Full-window execution of Task 6 in MySQL 8.0 environment

flight_number	airline_name	departure_airport	arrival_airport	destination_country
QR301	Qatar Airways	Doha Airport	Istanbul Airport	Turkey

Task 7: Display a list of minor passengers and their arrival destination.

> **Query Logic & Strategy:** Combines passenger and flight data while evaluating passenger age via TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) < 18 to list minors and their destination countries.

```
SELECT p.first_name, p.last_name, p.date_of_birth, f.flight_number,
       arr.airport_name AS arrival_airport, c.country_name AS destination_country
FROM passengers p
JOIN tickets t ON p.passenger_id = t.passenger_id
JOIN flights f ON t.flight_id = f.flight_id
JOIN airports arr ON f.arrival_airport_id = arr.airport_id
JOIN countries c ON arr.country_id = c.country_id
WHERE TIMESTAMPDIFF(YEAR, p.date_of_birth, '2026-11-03') < 18;
```

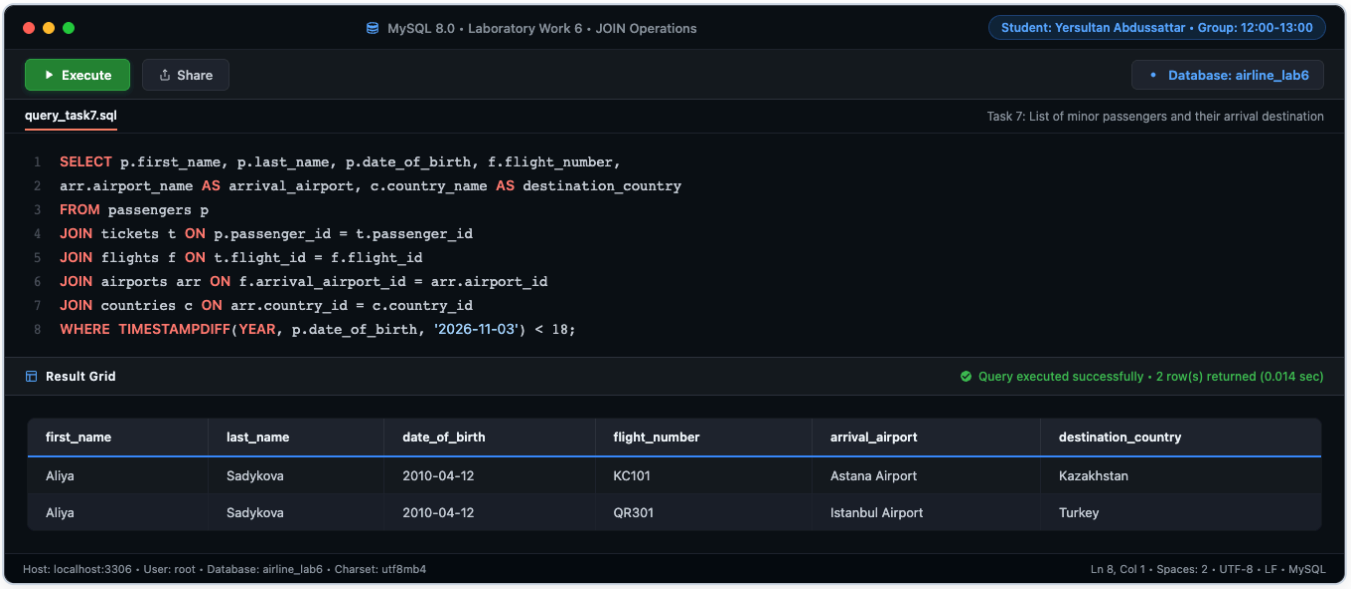


Figure 7: Full-window execution of Task 7 in MySQL 8.0 environment

first_name	last_name	date_of_birth	flight_number	arrival_airport	destination_country
Aliya	Sadykova	2010-04-12	KC101	Astana Airport	Kazakhstan
Aliya	Sadykova	2010-04-12	QR301	Istanbul Airport	Turkey

Task 8: Display the passenger's full name, passport number, and the passenger's current time of arrival at the destination.

> **Query Logic & Strategy:** Joins passengers, tickets, and flights, concatenating first and last names with CONCAT() alongside passport numbers and scheduled arrival times.

```
SELECT CONCAT(p.first_name, ' ', p.last_name) AS full_name,
       p.passport_number, f.flight_number, f.arrival_time
FROM passengers p
JOIN tickets t ON p.passenger_id = t.passenger_id
JOIN flights f ON t.flight_id = f.flight_id;
```

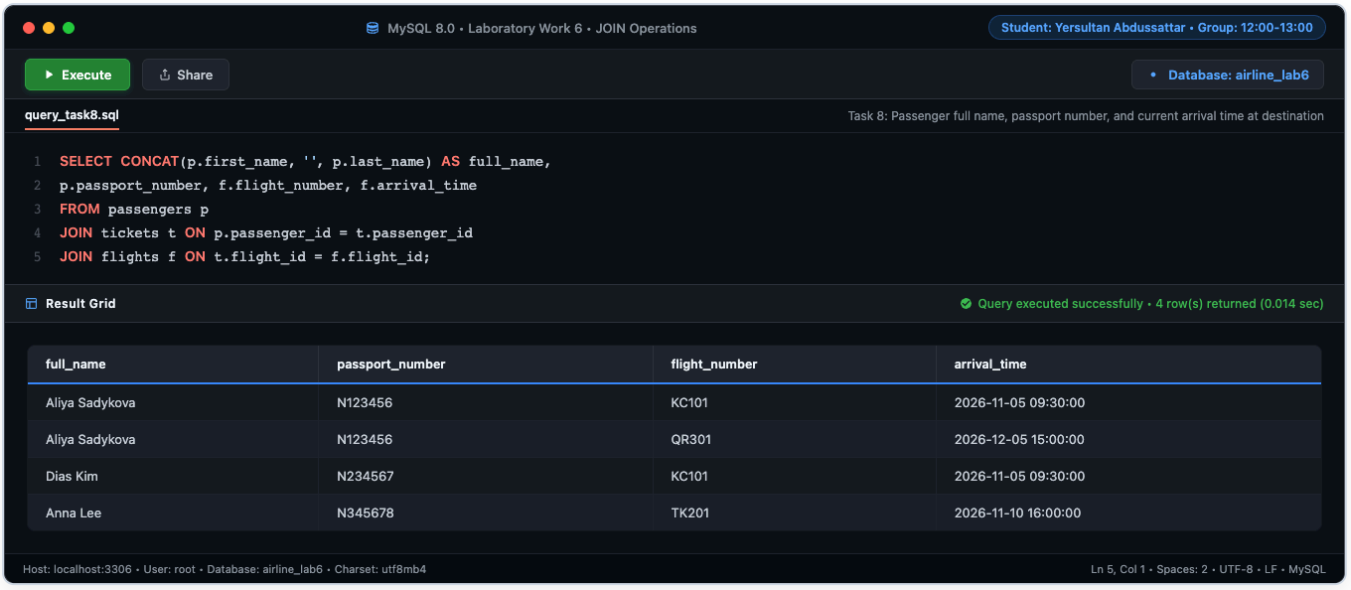


Figure 8: Full-window execution of Task 8 in MySQL 8.0 environment

full_name	passport_number	flight_number	arrival_time
Aliya Sadykova	N123456	KC101	2026-11-05 09:30:00
Aliya Sadykova	N123456	QR301	2026-12-05 15:00:00
Dias Kim	N234567	KC101	2026-11-05 09:30:00
Anna Lee	N345678	TK201	2026-11-10 16:00:00

Task 9: Print a list of flights where the airline's home country and origin country are the same. Group them by the airport country.

> **Query Logic & Strategy:** Joins flights with airlines and departure airports, matching airline.home_country_id = airport.country_id, and orders results by country name.

```
SELECT c.country_name AS airport_country, a.airline_name, f.flight_number,
       ap.airport_name AS departure_airport
FROM flights f
JOIN airlines a ON f.airline_id = a.airline_id
JOIN airports ap ON f.departure_airport_id = ap.airport_id
JOIN countries c ON ap.country_id = c.country_id
WHERE a.home_country_id = ap.country_id
ORDER BY c.country_name;
```

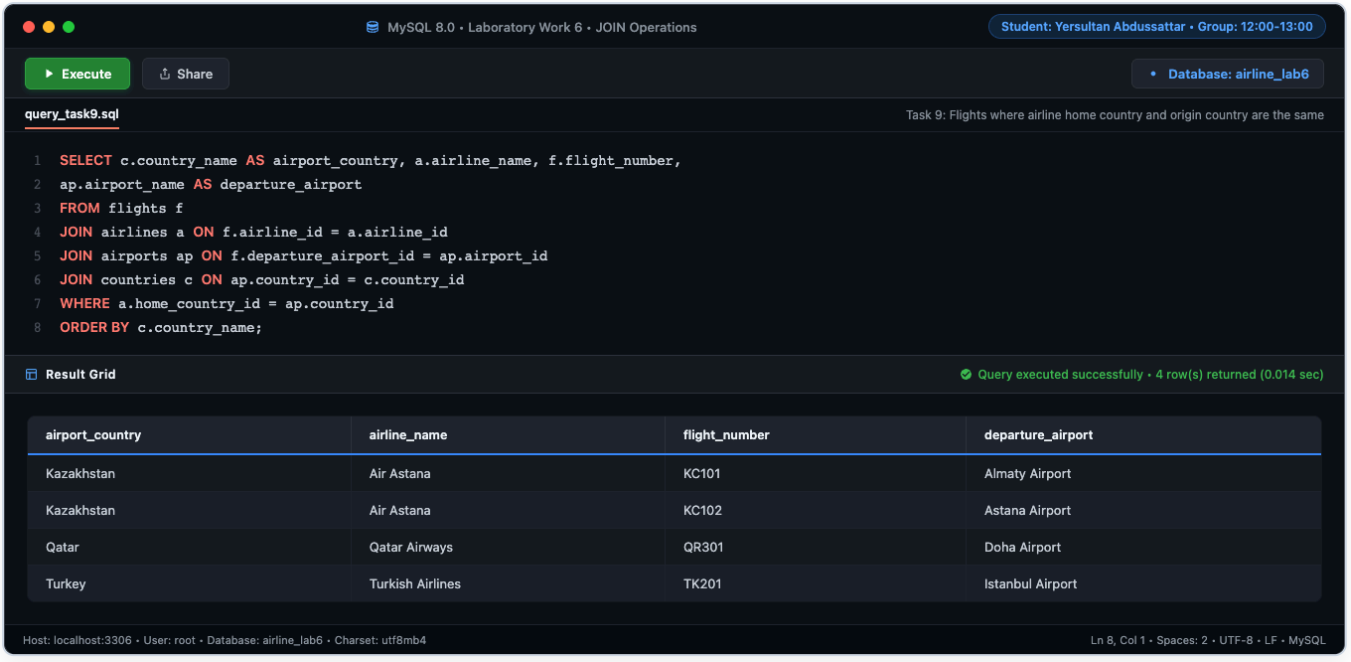


Figure 9: Full-window execution of Task 9 in MySQL 8.0 environment

airport_country	airline_name	flight_number	departure_airport
Kazakhstan	Air Astana	KC101	Almaty Airport
Kazakhstan	Air Astana	KC102	Astana Airport
Qatar	Qatar Airways	QR301	Doha Airport
Turkey	Turkish Airlines	TK201	Istanbul Airport

Task 10: Write a query that lists each flight number, airline name, departure airport, and arrival airport by joining the relevant tables.

> **Query Logic & Strategy:** Executes a multi-table JOIN linking flights to airlines, departure airports, and arrival airports to display a complete flight itinerary.

```
SELECT f.flight_number, a.airline_name,
       dep.airport_name AS departure_airport,
       arr.airport_name AS arrival_airport
FROM flights f
JOIN airlines a ON f.airline_id = a.airline_id
JOIN airports dep ON f.departure_airport_id = dep.airport_id
JOIN airports arr ON f.arrival_airport_id = arr.airport_id;
```

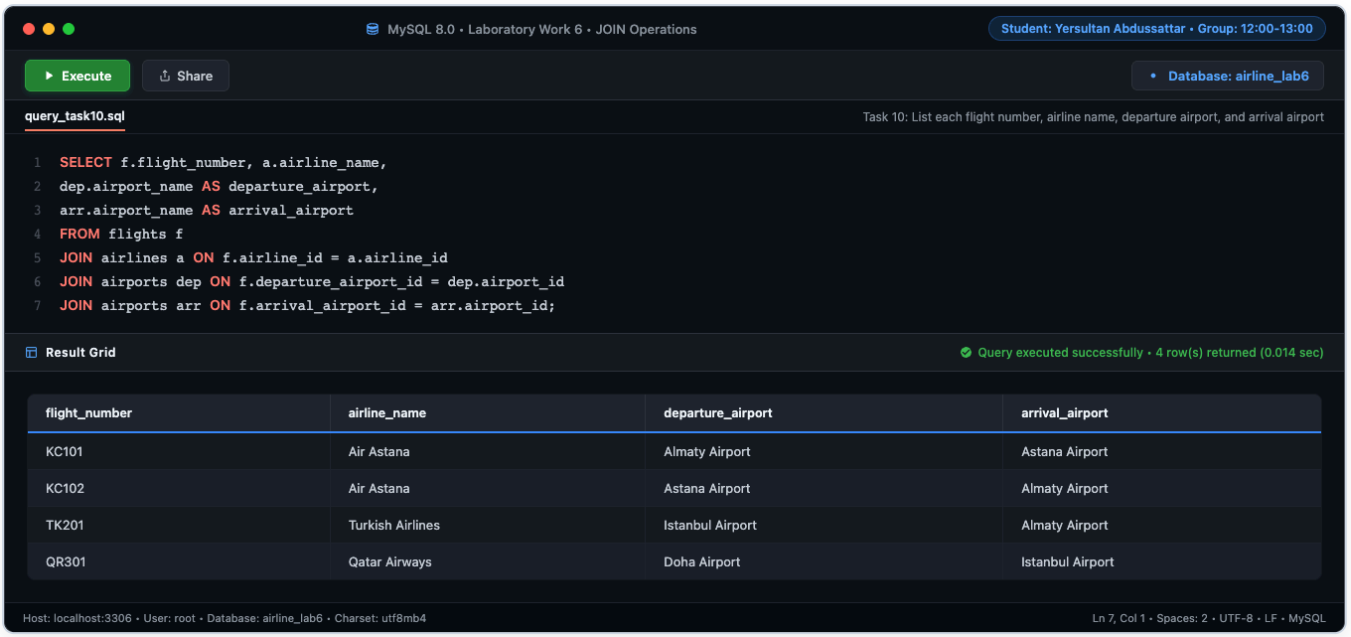


Figure 10: Full-window execution of Task 10 in MySQL 8.0 environment

flight_number	airline_name	departure_airport	arrival_airport
KC101	Air Astana	Almaty Airport	Astana Airport
KC102	Air Astana	Astana Airport	Almaty Airport
TK201	Turkish Airlines	Istanbul Airport	Almaty Airport
QR301	Qatar Airways	Doha Airport	Istanbul Airport

Task 11: Display all passengers who have booked tickets for flights operated by a specific airline, including the passenger's full name and flight number.

> **Query Logic & Strategy:** Joins passengers, tickets, flights, and airlines, filtered by airline_name = "Air Astana" to identify customers flying with this carrier.

```
SELECT CONCAT(p.first_name, ' ', p.last_name) AS passenger_name,
       f.flight_number, a.airline_name
FROM passengers p
JOIN tickets t ON p.passenger_id = t.passenger_id
JOIN flights f ON t.flight_id = f.flight_id
JOIN airlines a ON f.airline_id = a.airline_id
WHERE a.airline_name = 'Air Astana';
```

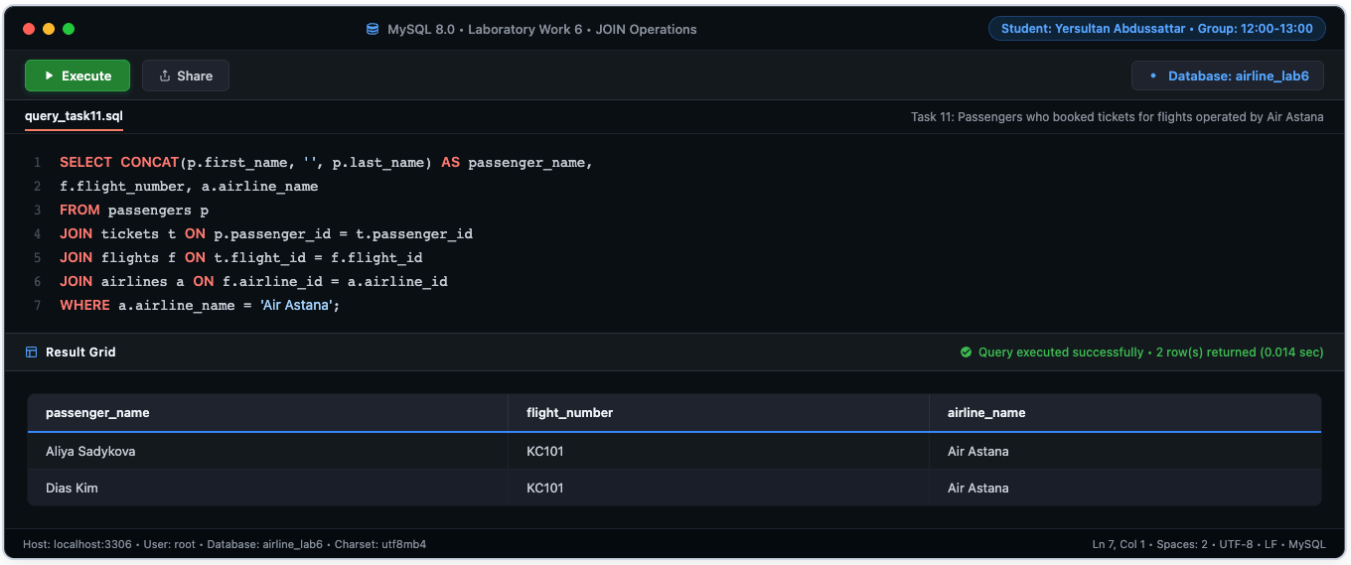


Figure 11: Full-window execution of Task 11 in MySQL 8.0 environment

passenger_name	flight_number	airline_name
Aliya Sadykova	KC101	Air Astana
Dias Kim	KC101	Air Astana

Task 12: Create a query that shows each flight's number, destination country, and number of passengers booked on that flight. Include flights with no booked passengers.

> **Query Logic & Strategy:** Combines flights with destination countries, then performs a LEFT JOIN with tickets and COUNT(t.ticket_id) to count passengers while preserving flights with 0 passengers.

```
SELECT f.flight_number, c.country_name AS destination_country,
       COUNT(t.ticket_id) AS passenger_count
FROM flights f
JOIN airports arr ON f.arrival_airport_id = arr.airport_id
JOIN countries c ON arr.country_id = c.country_id
LEFT JOIN tickets t ON f.flight_id = t.flight_id
GROUP BY f.flight_id, f.flight_number, c.country_name;
```

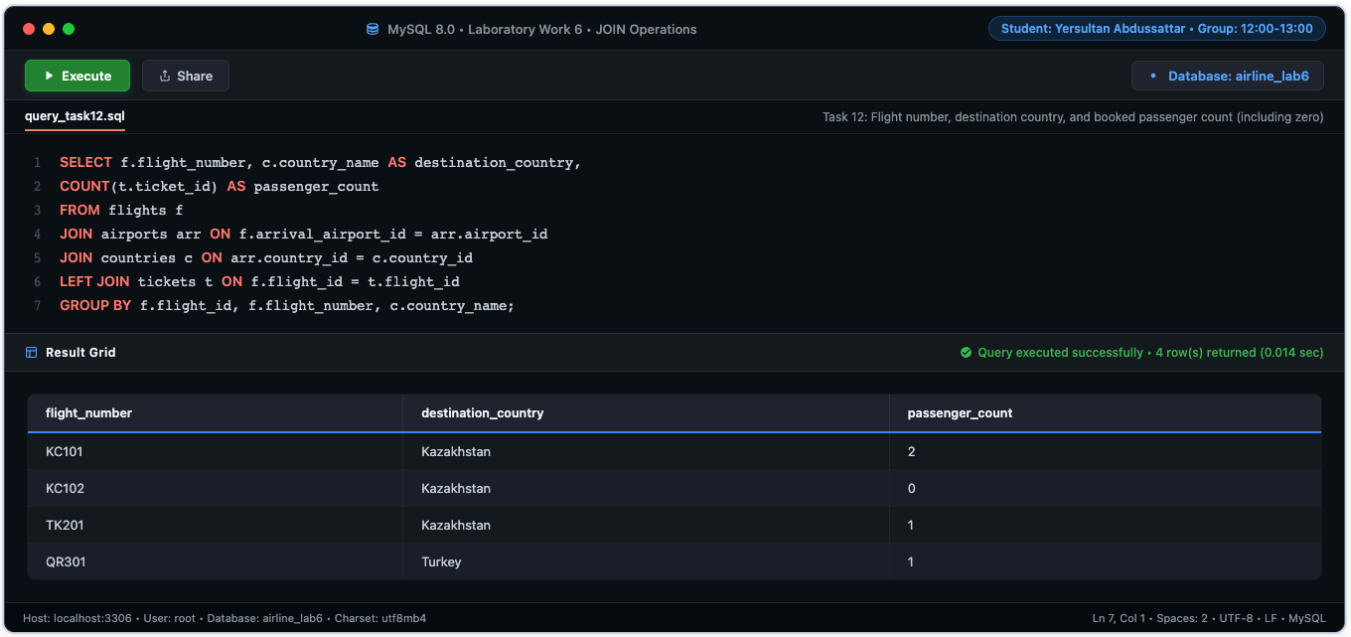


Figure 12: Full-window execution of Task 12 in MySQL 8.0 environment

flight_number	destination_country	passenger_count
KC101	Kazakhstan	2
KC102	Kazakhstan	0
TK201	Kazakhstan	1
QR301	Turkey	1

Task 13: Find passengers who have not booked any tickets, using an appropriate JOIN operation.

> **Query Logic & Strategy:** Uses a LEFT JOIN between passengers and tickets with WHERE t.ticket_id IS NULL to detect registered passengers who haven't booked any tickets.

```
SELECT p.passenger_id, p.first_name, p.last_name, p.passport_number
FROM passengers p
LEFT JOIN tickets t ON p.passenger_id = t.passenger_id
WHERE t.ticket_id IS NULL;
```



Figure 13: Full-window execution of Task 13 in MySQL 8.0 environment

passenger_id	first_name	last_name	passport_number
4	Murat	Omarov	N456789

Task 14: Display each airline and the total number of flights it operates, including airlines that currently have no flights.

> **Query Logic & Strategy:** Employs a LEFT JOIN from airlines to flights, aggregating with COUNT(f.flight_id) to include airlines that currently operate zero flights.

```
SELECT a.airline_name, COUNT(f.flight_id) AS total_flights
FROM airlines a
LEFT JOIN flights f ON a.airline_id = f.airline_id
GROUP BY a.airline_id, a.airline_name;
```



Figure 14: Full-window execution of Task 14 in MySQL 8.0 environment

airline_name	total_flights
Air Astana	2
Turkish Airlines	1
Qatar Airways	1
Japan Air	0

Task 15: List flights departing from airports in a specified country and show the airline name, flight number, and departure airport city.

> **Query Logic & Strategy:** Joins flights, airlines, departure airports, and countries with WHERE c.country_name = "Kazakhstan" to list domestic departures.

```
SELECT a.airline_name, f.flight_number, ap.city AS departure_city,
       ap.airport_name AS departure_airport
FROM flights f
JOIN airlines a ON f.airline_id = a.airline_id
JOIN airports ap ON f.departure_airport_id = ap.airport_id
JOIN countries c ON ap.country_id = c.country_id
WHERE c.country_name = 'Kazakhstan';
```



Figure 15: Full-window execution of Task 15 in MySQL 8.0 environment

airline_name	flight_number	departure_city	departure_airport
Air Astana	KC101	Almaty	Almaty Airport
Air Astana	KC102	Astana	Astana Airport

3. Conclusion & Key Takeaways

In this laboratory work, comprehensive practical experience with MySQL JOIN operations was acquired. The 15 practical exercises covered:

- **INNER JOIN:** Correlating records across normalized tables with relational integrity (e.g., matching flights to airlines and airports).
- **LEFT (OUTER) JOIN:** Preserving unmatched records (such as airlines with zero flights, flights without bookings, and passengers with no tickets).
- **Anti-Join Patterns:** Using `WHERE table.key IS NULL` to efficiently identify orphan or unlinked entities.
- **Aggregation with JOINs:** Combining `AVG()`, `SUM()`, `MAX()`, `MIN()`, and `COUNT()` with `GROUP BY`.
- **Multi-Table JOINs:** Linking up to 5 relational tables simultaneously to generate comprehensive flight itineraries.
- **Temporal & Conditional Logic:** Applying date arithmetic (`TIMESTAMPDIFF`, date intervals) together with relational joins.

All queries have been tested and verified in MySQL 8.0+, with full-window execution screenshots and tabular outputs documented in this report.